Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Evidence

Meaning in my words How would you explain evidence to a friend?	**How can you tell?** What reading supports the claim that the cart reached 4 m in 2 s?
Example or sketch How could two measurements support a claim that one cart accelerated more?	**Non-example and why** “I expected our cart to win.” Does that tell us which cart crossed first?

Inference

Meaning in my words How would you explain inference to a friend?	**How can you tell?** Wet tracks run from the door to a cart. Which part is your inference?
Example or sketch Two timed images show XJ closer to Earth. What can you reasonably infer?	**Non-example and why** “The second image reads 4 km.” Is that your inference or your reading?

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Model

Meaning in my words How would you explain model to a friend?	**How can you tell?** A cart moves 2 m each second. What are we assuming if we extend that pattern?
Example or sketch Where would your model put the cart one second after it passes 6 m going right?	**Non-example and why** The cart hits a cushion. Would you still extend the same pattern? Why?

Testable question

Meaning in my words How would you explain testable question to a friend?	**How can you tell?** What would you measure to answer “Does this cart speed up on the ramp?”
Example or sketch What question could two timed pictures of the same cart help you answer?	**Non-example and why** “Which cart looks coolest?” What would you change so the class can test it?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a three-sighting timeline. Label the starting position, change per second, and one future position. Use a dotted mark for the forecast.

Choose zero, direction, and scale.



My marks mean

On assignment: Evidence

SECTION E / BACK / FINAL MEETING ONLY Name _____

Three telescope fixes are at 2, 4 and 6 km at minutes 0, 1 and 2. The night crew has one more exposure.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Telescope scheduler**

Mark where to aim at minute 3. Tell the operator which dots were measured and which dot is your prediction.

☐ **Science broadcaster**

Give a two-sentence update to viewers. Point to a measured dot and say what the camera should see next if the pattern holds.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

Editor's check: Name an observation that would make you revise your prediction.